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Top Skills

Laser Cutting
Autodesk Fusion 360
Machine Learning

Certifications

From Excel to Tableau
Everybody's Introduction to Tableau (2022)
Tableau for Data Scientists
Corporate Finance: Environmental, Social, and Governance (ESG)

Emmanuel Sampson

Aspiring Solutions Engineer
Williamsburg, Virginia, United States

Experience

WEINERT Fiber Optics Inc

Engineer Intern
May 2025 - Present (6 months)
Williamsburg, Virginia, United States

Design and implement AI-powered computer vision systems in Python and C++ to create an automated fiber-optic end-face defect detector

Develop and optimize a real-time medical temperature monitoring and logging sensor

Transmission Testing

Polarization Alignment of Stress Rods for PM Fiber

Reconfigured and managed industrial robotics programming the user-interface and internal system of the Mach3 Mill to connect to a Levin Lathe

William & Mary

3 years 1 month

Laboratory Assistant

October 2023 - Present (2 years 1 month)
Williamsburg, Virginia, United States

Restructured the lab environment to optimize workflow.

I was responsible for managing a \$ 4,000 budget for acquiring new electronics and equipment necessary for the lab.

Used CAD Modeling to design an optical experiment in reflective optics of low-angle infrared reflectors.

Assisted research teams by operating classical and electron microscopes and organizing experimental data.

Designed, developed & conducted reflective-optics experiments to measure infrared light reflectivity at shallow angles

Engineered custom electronic devices for data-acquisition solutions

Restructured laboratory layout for optimal usage

Managed & operated advanced instrumentation: Hitachi S-4700 FE-SEM, Phenom SEM, Bruker Dimension ICON AFM, HIROX KH-7700, Renishaw inVia Raman, Dektak 3 ST, and Profilm3D surface profilers

Assembled TRIFT 4 time-of-flight mass spectrometry instrument for high-resolution surface analysis.

Marketing Communications Assistant

January 2024 - September 2025 (1 year 9 months)

Photographed key events such as research symposiums and promotional showcases

Designed a website for automatic photo editing to outmatch the upload and editing speed of Adobe Lightroom without sacrificing photo quality or editing capabilities.

Participated in a weekly think-tank-like environment, consistently pitching new ideas for effective marketing.

Designed and constructed a human-sized prop microscope for additional creative photography.

Human Resources Media Specialist

October 2022 - June 2024 (1 year 9 months)

Williamsburg, Virginia, United States

Create creative onboarding, tutorial, and instructional videos for both new and old employees to be fully prepared for the work environment.

Managed and indexed a dataset of 5000+ files for optimal and efficient use.

Student Makerspace Engineer

October 2022 - October 2023 (1 year 1 month)

Developed the 3d Scanning lab, which allowed anthropology students to document and preserve artifacts virtually.

Fulfilled over 1,000 service requests through the Microsoft Jira system for different unique machines such as LazerCutters, 3D Printers, and CNC Mills
Trained and certified over 200 students in the safe operation of advanced manufacturing equipment, providing expert advice for research and academic projects.

Designed and led a multi-level robotics workshop series, successfully accommodating participants from beginner to advanced skill levels.

Created on-demand educational tools using a QR code system, allowing users to instantly access machine information and tutorials.

Collaborated with student organizations to support outreach initiatives and promote inclusive maker education.

Maintained and repaired a fleet of 19 3D printers (Makerbot, Airwolf), a 150W CO2 Boss Laser cutter, and 8 sewing machines.

Proficiently operated and trained users on design software (Fusion 360, Apex) and fabrication hardware.

Produced a precision CNC-routed mold for a visiting professor's art exhibition, delivering a critical component for a gallery display.

Two Beagles Studio LLC

Production Assistant

March 2023 - July 2025 (2 years 5 months)

Wrote and developed intellectual property such as novels, screenplays, and advertisements.

In addition to providing commentary and ideas on projects for the head producer. I wrote and edited short scientific novels that required fact-based, intensive research, covering topics ranging from psychological analysis to technological trends and innovations.

Ideapress Publishing

Freelance Writer

January 2023 - June 2025 (2 years 6 months)

Wrote book activity guides and reflection activities.

CultureWorks Richmond

Film Producer

March 2022 - September 2024 (2 years 7 months)

Richmond, Virginia, United States

Filmed and edited documentaries on "The Modern Representations of Culture" in Richmond and the tri-cities of Virginia

Conducted hour-long interviews and research on local residents in the tri-cities.

The Virginian-Pilot

Photojournalist

May 2024 - June 2024 (2 months)

Norfolk, Virginia, United States

I shadowed every photographer of the Virginia-Pilot and pitched my own article on the technological intersection of Colonial Williamsburg, detailing ways the historic living museum hides the work they do behind the scenes.

Build RVA / Build Forward Foundation

Makerspace Engineer Intern

May 2023 - July 2023 (3 months)

Richmond, Virginia, United States

Managed laser cutting devices, woodworking stations, and helped in the reconstruction of a Voron 3D printer.

Constructed an electric motorized bike from scratch.

Interacted with startup engineering businesses.

JCPenney Portraits

Photographer

August 2021 - August 2022 (1 year 1 month)

Colonial Heights, Virginia, United States

I managed a high-speed environment that required effective customer engagement and advanced organizational skills.

In this role, I was recognized as the "Top Sales Associate" for two consecutive months.

Colonial Heights Public Schools

Elementary School Tutor

November 2021 - May 2022 (7 months)

Colonial Heights, Virginia, United States

I was the lead tutor and placed in charge of managing groups of elementary students to ensure they could engage with their assignments and understand their classes, regardless of the subject.

Private Tutoring

Private Tutor

September 2018 - November 2021 (3 years 3 months)

Colonial Heights City, Virginia, United States

I tutored math and science. During the time of the pandemic, I created fun educational videos that allowed my students to engage remotely.

Education

William & Mary

Bachelor's degree, Engineering Physics/Applied Physics & Philosophy · (August 2022 - May 2026)

UCLA School of Theater, Film & Television

Cinematography and Film/Video Production · (June 2024 - August 2024)

Colonial Heights High School

High School Diploma · (September 2018 - June 2022)